User Experience Design Methodologies for Building User Interfaces for Teleround System for Intensive Care Units

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Abstract

Integrating design methods with telehealth intervention development offers an opportunity to understand user needs and to address potential barriers to using new digital tools.

Objective: This study aimed to apply user experience (UX) design methodologies for the construction of interfaces with users of the Mangará Digital Telemedicine platform, specifically for carrying out Telerounds with Intensive Care Units (ICUs) in remote locations.

Methods: This is a methodological study for developing a telemedicine platform. We used the Double Diamond Model to guide the design process. This method consists of four stages (Discover, Define, Develop, and Deliver), which include the following steps: 1. Construction of the initial vision of the Product Discovery journey through the CSD Matrix (Certainties, Assumptions, and Doubts); 2. Desk Research (DR); 3. Rapid Ethnographic research; 4. Comparison with previous projects using the methodology of "benchmarking"; 5. Creation of personas and empathy maps 6. Situational Diagnosis of ICUs. 7. User journey mapping; 8. Initial creation of the digital product using the Sitemap and UserFlow tools. 9. Construction of the usability of wireframes according to Nielsen's heuristics.

Results: The use of UX methodologies for the design of the user interface was necessary for the construction of the Mangará Digital Telerounds platform since it allowed the development of a friendly platform that meets the expectations and needs of its users.

Conclusion: This work demonstrates the importance of using UX methodologies to design the interfaces of electronic medical record systems in high-demand and complex environments such as ICUs.

INTRODUCTION

Telehealth technologies in intensive care (IC) settings aim to provide care to critically ill patients and promote significant training of remote intensive care unit (ICU) teams. Thus, it has contributed to decreased mortality and length of stay, especially in units with difficulties hiring a specialized team [1-5].

Despite the enormous progress in using ICTs (information and communication technologies) in health, building electronic medical record systems with user-friendly interfaces in IC environments remains an important challenge[1, 3, 5, 6]. Integrating user experience (UX) design methods with telehealth intervention development offers an opportunity to understand user needs and the potential impact of new features. It can help address potential barriers to using new digital tools.

Although UX methods have been successfully employed in developing innovative telehealth interventions in different clinical areas, these interventions have predominantly been developed and implemented in high-income countries[7-19]. In Latin America, the application of UX principles in telehealth interventions is relatively unexplored, with only a few articles implementing these principles and none explicitly mentioning the utilization of the double diamond model to develop teleround software[13, 20, 21].

This work aims to apply UX methodologies to define the interface of the teleround recording system - Mangará Digital - used by specialists in IC in daily telerounds with remote ICUs.

METHODS

1.Study design

We conducted a methodological study using a human-centered design approach to design an interface for the Teleround recording system (Mangará Digital) from February 2022 to May 2022. This project was approved by the research ethics board at the Hospital Sírio-Libanês (Number: 6.215.082).

2. Setting

The project of Qualification of Care in IC through Telemedicine (NUP 25000.177384/2021-11) of the Institutional Development Program of the Unified Health System (Proadi-SUS) has the objective of providing, through the expertise of specialists in IC from health entities of recognized excellence, support to medical and multidisciplinary teams in the care of patients in the participating ICUs through telemedicine. Currently, Hospital Sírio-Libanês (HSL) supports 16 (sixteen) ICUs located in the states of Pará, Piauí, Maranhão, and Acre.

3 Design team

A design team consisted of a UX designer, an assistant with knowledge of customer service and communication processes, a nurse with expertise in IC, a project manager with experience in digital health, and a developer. The involvement of each member is outlined in Table 1.

4. Design Process

Double diamond is a design thinking methodology that offers a structured approach to solving complex problems by dividing the process into four stages: Discover, Define, Develop, and Deliver [22]. In the Discover and Define stages, the team works to understand the problem by researching and gathering information to develop a clear problem statement. The Develop and Deliver stages involve ideation, prototyping, and testing to generate and refine diverse ideas until a practical and effective solution is developed. Table 1 provides an overview of the double diamond steps.

4.1 Discover

4.1.1 Step 1- CSD Matrix (Certainties, Suppositions and Doubts)

A meeting was conducted in February 2022 between design team members to construct the CSD Matrix. The theme of the matrix was teleround. We used the Miro platform to collect and organize the data. Each idea was categorized as a certainty, supposition, or doubt. Certainties were based on the approved work plan and the team's previous experiences in digital health. Suppositions were derived from relevant literature and knowledge in IC, focusing on feasibility and applicability within the work plan and initial discussions. Doubts were formulated to address the specificities of locations, users, and technological possibilities to be explored.

4.1.2 Step 2- Desk Research (DR)

First, a meeting was conducted between design team members to define the research topics based on assumptions and doubts raised in the previous step. A literature search was conducted between the 15th and 25th of March in research databases (Research Gate, SciELO Brazil, Critical Care Nursing Quarterly, JMIR Publications) and websites (Google) using the following keywords: engagement, ICU, telemedicine, teleround, user feedback, integration, health system, data sharing and communication. Searches were limited to English and Portuguese languages.

Methodological studies, case studies, and interventions related to digital products that addressed the following topics were included: user engagement, technological structure, communication strategies, multidisciplinary teams, data collection, and sharing. The inclusion of articles was not restricted to the digital health area; thus, studies from other areas that were compatible with the research topics and those that described new technologies were included.

We excluded studies that applied design approaches, but the brief descriptions of their design work needed to be more comprehensive to meet the proposed objective.

The collected data were registered in Excel. Finally, a discussion group was performed during a meeting to discuss the findings and seek insights to support the project.

4.1.3 Step 3- Rapid Ethnography

We used rapid ethnography in ICUs from two hospitals (a public hospital and a private hospital) to understand the behavior patterns of the multidisciplinary team, ICU routines, and differences and similarities between the public and private services[23]. The ICUs were selected for convenience. The UX designer performed two face-to-face visits (5 hours each)

At each site, the UX designer conducted participant observations of key settings and semistructured interviews of staff members to understand the context (number of beds, type of patients, preparation for the rounds, registration of patient data, composition of the team) processes (routine of each professional, work schedule, digitalization, collection of indicators, visualization of exams, recording of rounds, use of a standardized checklist, existence of Standard Operating Procedures (SOP), and interpersonal dynamics (multidisciplinary team) of the ICU. The data collected were manually recorded and digitized in Notion.

4.1.4 Step 4- Benchmarking

We applied semistructured interviews via video with two project managers involved in implementing telemedicine methodologies within Brazilian public ICUs. The interview included the following topics: structure of the project (platforms used, situational diagnosis, and video communication technologies), processes (agenda management, standardization of care, routines with the ICUs), and human resources (users' adherence, interaction with technology and definition of roles).

4.2 Define

4.2.1 Step 5- Personas and Empathy Maps

Two empathy maps were developed, one representing medical users and another representing nursing users. The empathy map was divided into six sections: 1. What does the user think and feel? ; 2. What the user sees? ; 3. What the user hears? ; 4. What does the user say and do? ; 5. Pains; and 6. Gains. The points raised in each session of the map were based on knowledge acquired in the previous stages of the methodology. The UX designer and the nurse participated in constructing the empathy maps in a 2-hour video call meeting using the Miro platform.

4.2.2 Step 6- Situational Diagnosis

We conducted a situational diagnosis for each participating ICU in the project. A questionnaire (Google Forms) developed by team members was sent to each ICU coordinator. It was organized into five sections: 1. General Information about the hospital (hospital's size, availability of exams and medical devices on-site); 2. The infrastructure of the ICU (availability of X-ray and ultrasound machines at the bedside, level of care provided to COVID-19 patients, and usage of electronic and/or physical medical records); 3. Human resources (size, workload, and diversity of the ICU care team); 4. Workflows (availability of protocols and daily census); 5. ICU Indicators.

A descriptive analysis was performed for each of the variables collected.

4.3 Develop (ideation)

4.3.1 Step 7- User journey mapping

The development of the user journey involved two stages. The initial journey was crafted in the first stage through collaborative brainstorming sessions conducted during two meetings. In the second stage, a detailed user journey was built. Journey map outputs were reviewed and discussed during six virtual meetings between team members, a project coordinator, and a data specialist.

4.3.2 Step 8- Sitemap and UserFlow

The UX designer created the sitemap and user flow using a Figma platform based on the results obtained in the ideate stage. The main modules, such as the initial screen with the "cards" (block of summarized information of the patients) of each patient, medical records, page for the insertion of indicators, and list of the registration of patients and hospitals, were generated based on usability concepts and the application of Nielsen's heuristic analysis.

4.4.3 Step 9- Wireframes construction

The creation process of the wireframe involved the development of initial prototypes and the proposal of the interface structure. It was accomplished using the Figma tool, and the UX/UI designer guided it. Additionally, a project specialist with expertise in data and digital health and a specialist in health intelligence and technology provided valuable insights during this step. The Wireframe construction phase lasted approximately four weeks and consisted of four synchronous meetings to discuss action functionalities and asynchronous activities for production.

Concurrently with the Wireframe construction, the Platform Development stage commenced with a focus on defining the databases. Two programmers were assigned to develop the screens, and the minimum viable product (MVP) was delivered within three months.

4.4 Deliver

4.4.1 Step 10 - Testing and validation

Finally, we used Nielsen's heuristics [6] to build the usability of wireframes. Meetings were held with the TeleUTI project team (doctors, nurses, and physiotherapists) to discuss the visualization of the screens and technical and functional details of the solution. The suggestions were registered on the Figma platform. It was impossible to carry out the usability test due to the time and schedule agreed upon in the work plan.

RESULTS

The design process has enabled the creation of the Mangará Digital Telerounds platform. The system went live on June 21, 2022. As of October 29, 2022, 877 telerounds were performed, in which 974 distinct patients were seen. The results obtained in each step of the Double Diamond are described below.

1. Step 1 - CSD Matrix

We identified 11 certainties, 11 suppositions, and six doubts. A detailed description of certainties, assumptions, and doubts can be found in Figure 1.

2. Step 2- Desk Research

Nine articles addressing the points listed in the CSD matrix ("engagement," "feedback," "infrastructure," "collect and sharing information," and "people resources") were selected[3, 6, 24-29]. Additionally, a book was selected under the "engagement" category, and three websites belonging to technological equipment suppliers were included within the "infrastructure" category (Supplementary material l).

Based on the literature selected, we considered the following key points during the project's development:

1. Participation of the ICU physicians of the sites from the implementation to improve the engagement of the medical area.
2. Teleround should be performed at the bedside to build trust between teams;
3. Implementing mechanisms for continuous evaluation of telerounds by the multidisciplinary team;
4. Sharing of information synchronously;
5. Using a structured and standardized instrument to conduct the teleround;
6. Using tablets requires mapping the main technological and structural barriers regarding using a less complex technical solution.

3. Step 3- Rapid ethnography

A comparison between a private and a public ICU in terms of their structure, processes, and human resources is presented in Supplementary Material 2. Despite the implementation of digital systems in both ICUs, there is a distinction in their medical record practices. The public ICU employs a hybrid approach, combining paper-based, scanned, and electronic medical records, while the private ICU exclusively utilizes electronic medical records. Another notable difference can be observed in the monitoring of indicators. The private ICU adheres to a structured agenda for collecting indicators, while the public ICU does not have a defined schedule.

4. Step 4 - Benchmarking

We interviewed professionals who participated in the TeleICU Covid and TeleICU Neonatal projects. Our findings revealed a marked difference in the planning stages. While the TeleICU Neonatal project had a well-defined plan, the TeleICU Covid project lacked comprehensive preplanning. Consequently, the execution of the Neonatal TeleICU project was notably more efficient and yielded superior outcomes. Implementing structured rounds and checklists and utilizing an appropriate platform was pivotal in organizing and collecting data and performance indicators. Furthermore, adopting a predefined agenda with fixed schedules proved highly beneficial in achieving favorable results.

5. Step 5- Personas and Empathy Maps

The empathy map identified the main pains of the ICU professionals: lack of time, overload, lack of agility and process standardization, the difficulty of viewing the evolution data of each patient, and the large number of manual processes still on paper (Figure 2).

6. Step 6- Situational Diagnosis

Of the 12 ICUs participating in the project, 10 (83.3%) responded. We identified several obstacles during the study, including challenges with internet connectivity (>50% of hospitals experience internet disruptions more than three times per month, and 20% lack broadband internet altogether), a deficiency in crucial ICUs, and inadequate staffing. A comprehensive examination of ICU infrastructure, workflows, human resources, and performance metrics can be found in Supplementary Material 3.

7. Step 7- User journey mapping

The user journey includes three main stages: preround, round, and postround. The first stage consists of the team meeting and entering the platform through the tablet.

The second stage is the teleround when the multidisciplinary teams of the remote ICUs discuss each patient with the teams from the TeleICU project. In the third stage, the user evaluates the teleround via the NPS (Net Promoter Score) (Figure 3).

This construction was essential to map the platform's functionalities according to each journey stage. Through actions taken by the user, we mapped what should be visualized on the interface and the flow of information that each new functionality should contain (Figure 3).

8. Step 8- Sitemap and UserFlow

In this stage, the main modules, such as the initial screen with the "cards" (block of summarized information of the patients) of each patient, the medical record, the page for the insertion of indicators, and the list of the registration of patients and hospitals, were generated based on usability concepts and the application of Nielsen's heuristic analysis. The initial ideas for the teleround registration screen were also produced with the creation of wireframes. The development process was started after the ideation and prototyping process with the wireframes (Supplementary material 4).

9. Step 9- Wireframes construction

We incorporated visually appealing and informative cards displaying key data such as the patient's age, name, duration of hospitalization, and any known allergies. These cards serve as a concise and easily digestible summary of the patient's essential details (Supplementary material 5).

Furthermore, the initial screen provides the healthcare team with a comprehensive map of the ICU's bed layout, facilitating efficient bed management and patient assignment. This visual representation enables informed decision-making regarding patient allocation and resource utilization.

Additionally, the rounding screen offers a wealth of vital information for healthcare professionals. It presents the patient's problem list, offering insights into the patient's specific health issues. Moreover, it provides:

- A comprehensive compilation of all the necessary data gathered from the daily checklist.
- Ensuring a comprehensive overview of the patient's condition.
- Treatment progress.
- Care plan**.**

10. Step 10 -Testing and validation

We identified areas for improvement in the platform; issues arose with the database during the user registration process, causing problems. The bed map feature showed that including the specific bed information proved challenging, as many users should have mentioned it. Suggestions were made to enhance the video tools by incorporating a chat feature, as audio and video problems frequently hindered user communication. These suggestions were among various other improvements identified (Supplementary material 6).

DISCUSSION

Although building electronic medical record systems for complex and critical environments such as ICUs is a challenge, this study shows how a human-centered design process can produce a usable teleround platform for multiple team members involved in IC. We employed a UX design methodology called Double Diamond, which was crucial for identifying challenges and opportunities in the development of the Mangará Digital Telerounds platform, proving to be an important tool for increasing the potential implementation of health technologies.

In the literature, some studies for developing online programs to deliver telehealth have also demonstrated the benefits of designing and delivering interventions following the double diamond process[8, 9]. To the best of our knowledge, this is the first study that clearly states implementing the double diamond model to develop teleround software in the IC. In addition, this study applies this design process in a Brazilian real-world scenario where the budget is limited for telehealth. Finally, we used a double diamond model in the early stages of design; situated methods can assist in determining the real needs of users and help anticipate responses to potential barriers. Thus, we contribute to practice by describing characteristics that may be relevant when applying design methodologies in healthcare.

An important limitation of the study is that the results observed about the public ICU during the rapid ethnographic research may not be generalized to all the ICUs participating in the project since the ICUs included in the study belonged to the State of São Paulo. This was because we did not know the ICUs that would participate in the project at that time.

Another significant limitation is that design methods most often rely on studying a relatively small sample in depth; this approach is prone to a range of sampling biases. By default, we are not talking to the right people, as not everybody can join the sessions; some specific groups will not be represented at all, whereas other groups are

A similar study with other providers in different settings may yield a different prototype. However, this is the first study in this context, and additional research is needed to understand how this intervention could benefit other ICU contexts.

The methods described in our study can be used as a template for future design considerations specific to developing digital tools. Future studies could explore the potential effects of particular design and implementation modifications on clinical outcomes. The design and implementation process for telehealth interventions is rarely linear and requires repeated adaptation to the changing needs of patients and clinicians.

This work demonstrates the importance of using UX methodologies to design the interfaces of electronic medical record systems in high-demand and complex environments such as ICUs.

Declarations

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AUTHORS' CONTRIBUTIONS

AFM, SVC and SDG participated in the conception and design of the study. AFM, LMS, ICS, SDG, WAS, and BFL contributed to the acquisition, analysis and interpretation of data. FRM, AFM, LMS, BFL, and SDG revised the article critically. All authors participated in the preparation of the manuscript and agreed to the submitted version of the paper.

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AVAILABILITY OF DATA AND MATERIALS

The data will be made available upon reasonable request.

ETHICS APPROVAL AND CONSENT TO PARTICIPATE

The study was approved by the Research Ethics Committee of Hospital Sírio-Libanês (number. 6.215.082) with a waiver of informed consent and conducted in accordance with the Declaration of Helsinki.

CONSENT FOR PUBLICATION

Not applicable.

COMPETING INTERESTS

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

1. Macedo BR, Garcia MVF, Garcia ML, Volpe M, Sousa MLA, Amaral TF, Gutierrez MA, Barbosa AP, Scudeller PG, Caruso P *et al*: **Implementation of Tele-ICU during the COVID-19 pandemic**. *J Bras Pneumol* 2021, **47**(2):e20200545.
2. Watanabe T, Ohsugi K, Suminaga Y, Somei M, Kikuyama K, Mori M, Maruo H, Kono N, Kotani T: **An evaluation of the impact of the implementation of the Tele-ICU: a retrospective observational study**. *J Intensive Care* 2023, **11**(1):9.

3. Guinemer C, Boeker M, Furstenau D, Poncette AS, Weiss B, Morgeli R, Balzer F: **Telemedicine in Intensive Care Units: Scoping Review**. *J Med Internet Res* 2021, **23**(11):e32264.
4. Becker CD, Fusaro MV, Al Aseri Z, Millerman K, Scurlock C: **Effects of Telemedicine ICU Intervention on Care Standardization and Patient Outcomes: An Observational Study**. *Crit Care Explor* 2020, **2**(7):e0165.
5. Chen J, Sun D, Yang W, Liu M, Zhang S, Peng J, Ren C: **Clinical and Economic Outcomes of Telemedicine Programs in the Intensive Care Unit: A Systematic Review and Meta-Analysis**. *J Intensive Care Med* 2018, **33**(7):383-393.
6. Ramnath VR, Hill L, Schultz J, Mandel J, Smith A, Holberg S, Horton LE, Malhotra A, Friedman LS: **Designing a critical care solution using in-person and telemedicine approaches in the US-Mexico border area during COVID-19**. *Health Policy Open* 2021, **2**:100051.
7. Asbjornsen RA, Wentzel J, Smedsrod ML, Hjelmesaeth J, Clark MM, Solberg Nes L, Van Gemert-Pijnen J: **Identifying Persuasive Design Principles and Behavior Change Techniques Supporting End User Values and Needs in eHealth Interventions for Long-Term Weight Loss Maintenance: Qualitative Study**. *J Med Internet Res* 2020, **22**(11):e22598.
8. Banbury A, Pedell S, Parkinson L, Byrne L: **Using the Double Diamond model to co-design a dementia caregivers telehealth peer support program**. *J Telemed Telecare* 2021, **27**(10):667-673.
9. Hardy A, Wojdecka A, West J, Matthews E, Golby C, Ward T, Lopez ND, Freeman D, Waller H, Kuipers E *et al*: **How Inclusive, User-Centered Design Research Can Improve Psychological Therapies for Psychosis: Development of SlowMo**. *JMIR Ment Health* 2018, **5**(4):e11222.
10. Garne Holm K, Brødsgaard A, Zachariassen G, Smith AC, Clemensen J: **Participatory design methods for the development of a clinical telehealth service for neonatal homecare**. *SAGE Open Med* 2017, **5**:2050312117731252.
11. Hobson EV, Baird WO, Partridge R, Cooper CL, Mawson S, Quinn A, Shaw PJ, Walsh T, Wolstenholme D, McDermott CJ: **The TiM system: developing a novel telehealth service to improve access to specialist care in motor neurone disease using user-centered design**. *Amyotroph Lateral Scler Frontotemporal Degener* 2018, **19**(5-6):351-361.
12. Nielsen C, Agerskov H, Bistrup C, Clemensen J: **User involvement in the development of a telehealth solution to improve the kidney transplantation process: A participatory design study**. *Health Informatics J* 2020, **26**(2):1237-1252.
13. Ozkaynak M, Sircar C, Frye O, Valdez R: **A Systematic Review of Design Workshops for Health Information Technologies**. *Informatics* 2021, **8**:34.
14. Frew M, Wyndham-West CM, Abrams HB: **Virtual care: a design research discovery and strategic futures model for the Canadian healthcare system**. *Design for Health* 2022, **6**(3):296-315.
15. Nesbitt K, Beleigoli A, Champion S, Gebremichael LG, Bulamu N, Du H, Tirimacco R, Clark RA: **Development and evaluation of a co-designed website for delivering interactive self-directed cardiac rehabilitation**. *Eur J Cardiovasc Nurs* 2023.
16. Villalba-Mora E, Ferre X, Perez-Rodriguez R, Moral C, Valdes-Aragones M, Sanchez-Sanchez A, Rodriguez-Manas L: **Home Monitoring System for Comprehensive Geriatric Assessment in Patient's Dwelling: System Design and UX Evaluation**. *Front Digit Health* 2021, **3**:659940.
17. Scanzera AC, Beversluis C, Potharazu AV, Bai P, Leifer A, Cole E, Du DY, Musick H, Chan RP: **Planning an artificial intelligence diabetic retinopathy screening program: a human-centered design approach**. *Frontiers in Medicine* 2023, **10**.
18. van Dormalen E: **Service Design for accessible rural healthcare in Finnish Lapland**. 2022.
19. Gerards T: **Tele-rehabilitation platform, a total knee arthroplasty prototype study**. 2018.
20. Dominguez-Rodriguez A, De La Rosa-Gómez A: **A Perspective on How User-Centered Design Could Improve the Impact of Self-Applied Psychological Interventions in Low- or Middle-Income Countries in Latin America**. *Front Digit Health* 2022, **4**:866155.
21. Nunes M, Teles AS, Farias D, Diniz C, Bastos VH, Teixeira S: **A Telemedicine Platform for Aphasia: Protocol for a Development and Usability Study**. *JMIR Res Protoc* 2022, **11**(11):e40603.
22. **The Double Diamond Design Process Model** [http://www.designcouncil.org.uk]
23. Ackerman S, Gleason N, Gonzales R: **Using Rapid Ethnography to Support the Design and Implementation of Health Information Technologies**. *Stud Health Technol Inform* 2015, **215**:14-27.
24. Compton M, Soper M, Reilly B, Gettle L, List R, Bailey M, Bruschwein H, Somerville L, Albon D: **A Feasibility Study of Urgent Implementation of Cystic Fibrosis Multidisciplinary Telemedicine Clinic in the Face of COVID-19 Pandemic: Single-Center Experience**. *Telemed J E Health* 2020, **26**(8):978-984.
25. Conroy I, Murray A, Kirrane F, Cullen L, Anglim P, O'Keeffe D: **Key requirements of a video-call system in a critical care department as discovered during the rapid development of a solution to address COVID-19 visitor restrictions**. *JAMIA Open* 2021, **4**(4):ooab091.
26. Larinkari S, Liisanantti JH, Ala-Laakkola T, Merilainen M, Kyngas H, Ala-Kokko T: **Identification of tele-ICU system requirements using a content validity assessment**. *Int J Med Inform* 2016, **86**:30-36.
27. Maran E, Matsuda LM, Marcon SS, Haddad MdCFL, Costa MAR, Magalhães AMMd: **ADAPTATION AND VALIDATION OF A MULTIDISCIPLINARY CHECKLIST FOR ROUNDS IN THE INTENSIVE CARE UNIT**. *Texto & Contexto - Enfermagem* 2022, **31**.
28. Sangoi K, Griebeler A, Aires M, Kinalski S: **ROUND MULTIPROFISSIONAL EM UNIDADE DE TERAPIA INTENSIVA: DISCUSSÃO PARA A IMPLANTAÇÃO DE PROCEDIMENTO OPERACIONAL PADRÃO**. In., edn.; 2020: 187-194.
29. Welsh C, Rincon T, Berman I, Bobich T, Brindise T, Davis T: **TeleICU Interdisciplinary Care Teams**. *Crit Care Nurs Clin North Am* 2021, **33**(4):459-470.

Table

Table 1. Overview of the double diamond stages and their main outcomes

Stages	Steps	Activities	When?	Participants	Data Collection	Data Organization	Data Analysis	Outputs/Outcomes
DISCOVER	CSD Matrix (Certainties, Suppositons and Doubts)	1 meeting (2 h)	February 2022	Members of design teams	Brainstorming	The data collected was recorded on the Miro platform	The data collected was classified into certainty, assumption or doubt about the project and its context.	Understanding the universe of the problem.
	Desk Research (DR)	Literature review + 2 synchronous meetings	March 2022	Members of design teams	Literature review in databases of scientific publications and websites + Group discussions on the results identified in the literature.	Data were be recorded in the Excel platform and structured in the Power Point program for further discussion of the results.	Content analysis was performed. The results of the studies were categorized according to the following topics: methods of engagement, user perception, technological structure, communication strategies, multidisciplinary team, data collection and sharing..	-Understanding of the main problems in digital health in Brazil and in the world. -Mapping about project experiences and technological solutions, whether in the health area or not, in Brazil and/or abroad, in order to generate insights that can help decision-making.
	Rapid Ethnographic Research	Face-to-face visits to an ICU of a private hospital (5 hours)	March 2022	-UX Designer (researcher)] -Private ICU staff members: 3 intensivist physicians, 2 nutritionists, 2 physiotherapists, 3 nurses. 2 residents and 1 psychologist).	Semistructured interviews + Participant observation	The collected data were registered manually in a physical form.	A comparative analysis of the ICUs was carried out in order to guarantee a better perspective of the daily life of an ICU and to build the best experience.	It was bringing the design team closer to the reality and processes in the ICU, as well as the behavioral patterns of the multidisciplinary team in the scenario, understanding the main triggers and insights of the users, allowing the construction of a journey that offers the best possible experience.
		Face-to-face visits to an ICU of a public hospital in a region with low economic conditions (5 hours)	March 2022	UX Designer (researcher) - Public ICU staff members: 1 Platonist doctor 1 speech therapist 1 physiotherapists 1 nurses 2 nursing technicians				

Table 1. Overview of the double diamond stages and their main outcomes (continuation)

	Steps	Activities	When?	Participants	Data Collection	Data Organization	Data Analysis	Outputs/Outcomes
	Benchmarking	Remote interviews	March 2022	- Members of design team -Coordinator of |Neonatal Tele-ICU project -Coordinator of Covid Tele-ICU project	Semistructured interviews	The collected data were recorded on the Notion platform	Comparative Analysis of project practices and experiences (in their different structures and scenarios) seeking points of convergence with the scope of the project and feasibility of adaptation and execution.	We are mapping the structure, execution and processes of similar digital health projects in order to understand the main barriers and facilities to support decision-making with a view to the best results and strategies.
DEFINE	Personas e Mapa de Empatia	Remote meeting (2 hours)	April 2022	Members of design team	Group discussions + brainstorming	The collected data were recorded on the Miro platform	The empathy map was built based on the survey of information in the previous stages, that is, in this stage the information has already been analyzed and will be organized	Understanding the context and behavioral aspects of physicians and nurses who work in public ICUs
	Situational Diagnosis	Questionnaires via the Survey platform	April 2022	Coordinators of the ICUs that are part of the project	self-administered questionnaire	The collected data was recorded on the survey's platform.	A descriptive analysis of the answers obtained in the forms was carried out. A comparative analysis of the data was also carried out with the aim of identifying differences in the resources available in each ICU. Finally, an analysis of the main initial barriers that could impact the execution and effectiveness of the project was carried out.	Knowing the structure and resources available in the ICUs, identifying problems and needs in order to anticipate predictions and solutions. They are building a user journey that can be applied in different scenarios (knowing their common patterns and difficulties).

Table 1. Overview of the double diamond stages and their main outcomes (continuantion)								
Stages	**Steps**	**Activities**	**When?**	**Participants**	**Data Collection**	**Data Organization**	**Data Analysis**	**Outputs/Outcomes**
DEVELOP	User Journey	Construction of the initial journey (2 synchronous meeting)	April 2022	Members of design team	Group discussions, Brainstorming	The collected data were recorded on the Miro platform	The User's Journey was built based on the collection of information in the previous stages, that is, in this stage the information has already been analyzed and is organized in a structured way. For data structuring, a map was systematized in vertical columns addressing the following topics: patient entry, ICU routines, case discussions	Architect the user journey of the TeleUTI project considering the assumptions and knowledge acquired in the previous steps.
		Detailed Journey Construction -6 synchronous meetings (1-3 hours)	April 2022	Members of design team + Stakeholders: Digital Health Project Coordinators, Developer, Data Specialist				
	Userflow + Sitemap	Creation of userflows and sitemap	April 2022	UX designer	-------------	Mapping of the main tasks and workflows that users perform on the product using the Figma Platform.	Hierarchical analysis of tasks, specifically for the purpose of developing our prototype.	Visual representation of tasks and paths users can take when interacting with the product
	Construction "Wireframe"	Meetings	April 2022	UX/UI designer (responsible for building the architecture in the tool), a project specialist expert in data and digital health and an expert in health intelligence.	Group discussions, Brainstorming	Figma platform	Systematization of the feedback obtained.	Visually structure the organization of platform elements and validate the feasibility of the planning defined in the previous steps
DELIVER (Test + Validation)	Testing and Validation of the User flow and the wireframes	Meetings	May 2022	UX Designer + TeleICU Project multidisciplinary team (doctors, nurses, physiotherapists)	Group discussions	The collected data will be registered on the Figma platform by the UX Designer	- Systematization of feedback obtained to identify opportunities for improvement.	Discuss screen views and technical and functional details

Figures

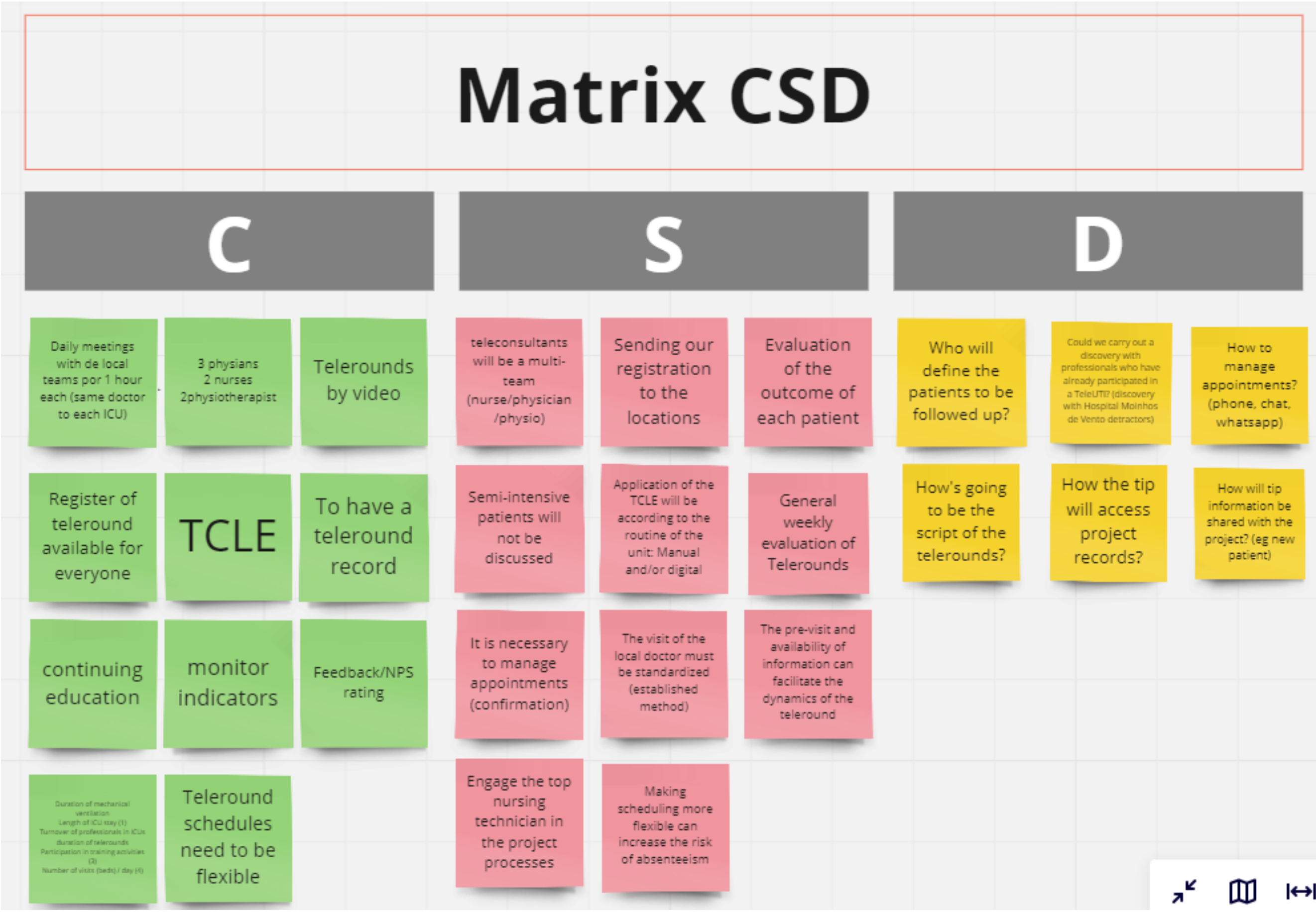

Figure 1

Matrix of Certainties, Assumptions and Doubts (CSD).

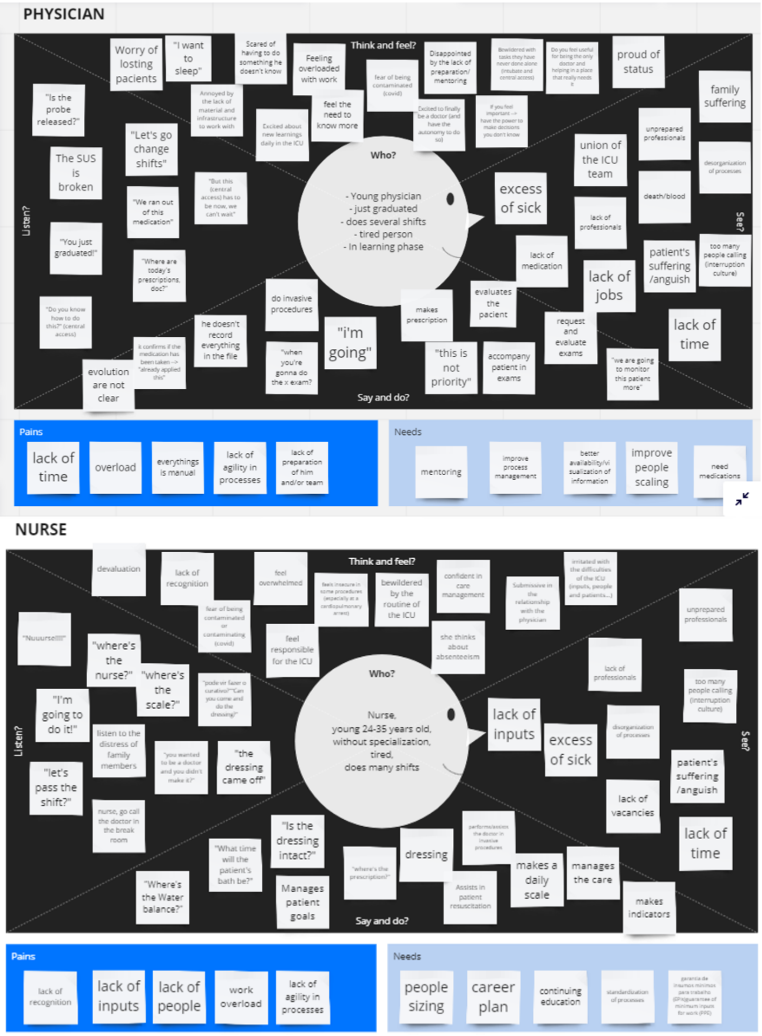

Figure 2

Personas and Empathy Maps of “Physician” persona and “Nurse” persona.

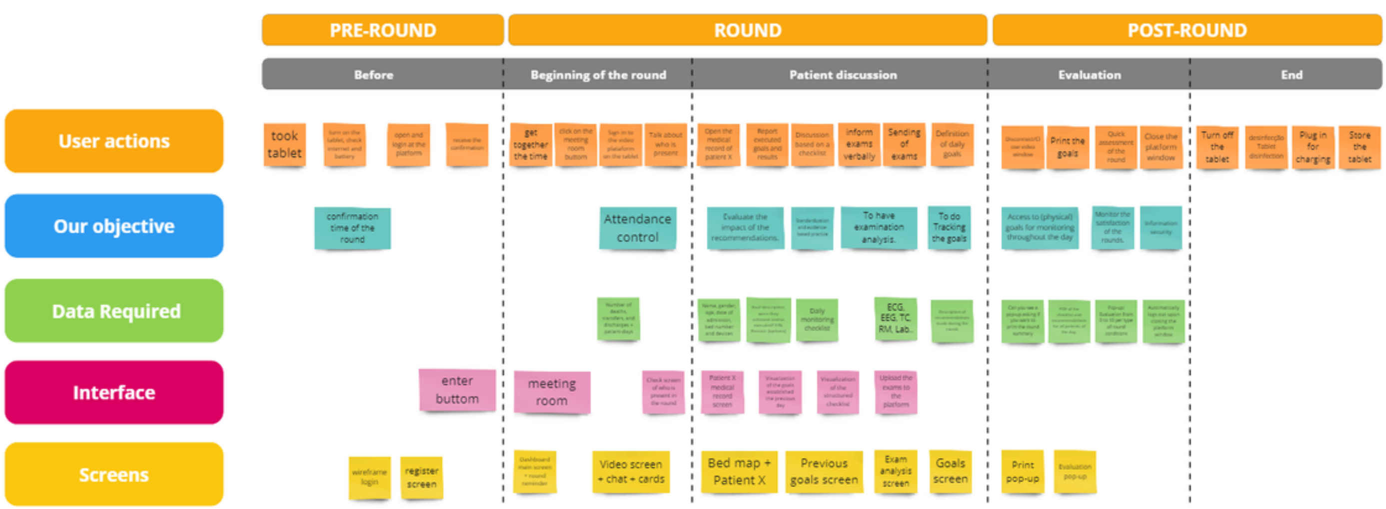

Figure 3

User Journey

Supplementary Files

This is a list of supplementary files associated with this preprint. Click to download.

- Supplementarymaterial1.docx
- Supplementarymaterial2.docx
- Supplementarymaterial3.docx
- Supplementarymaterial4.docx
- Supplementarymaterial5.docx
- Supplementarymaterial6.docx